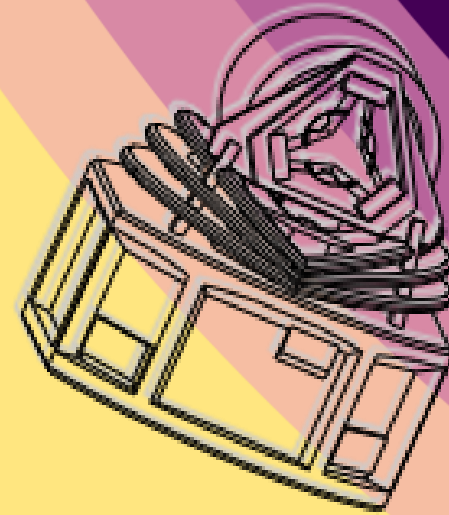


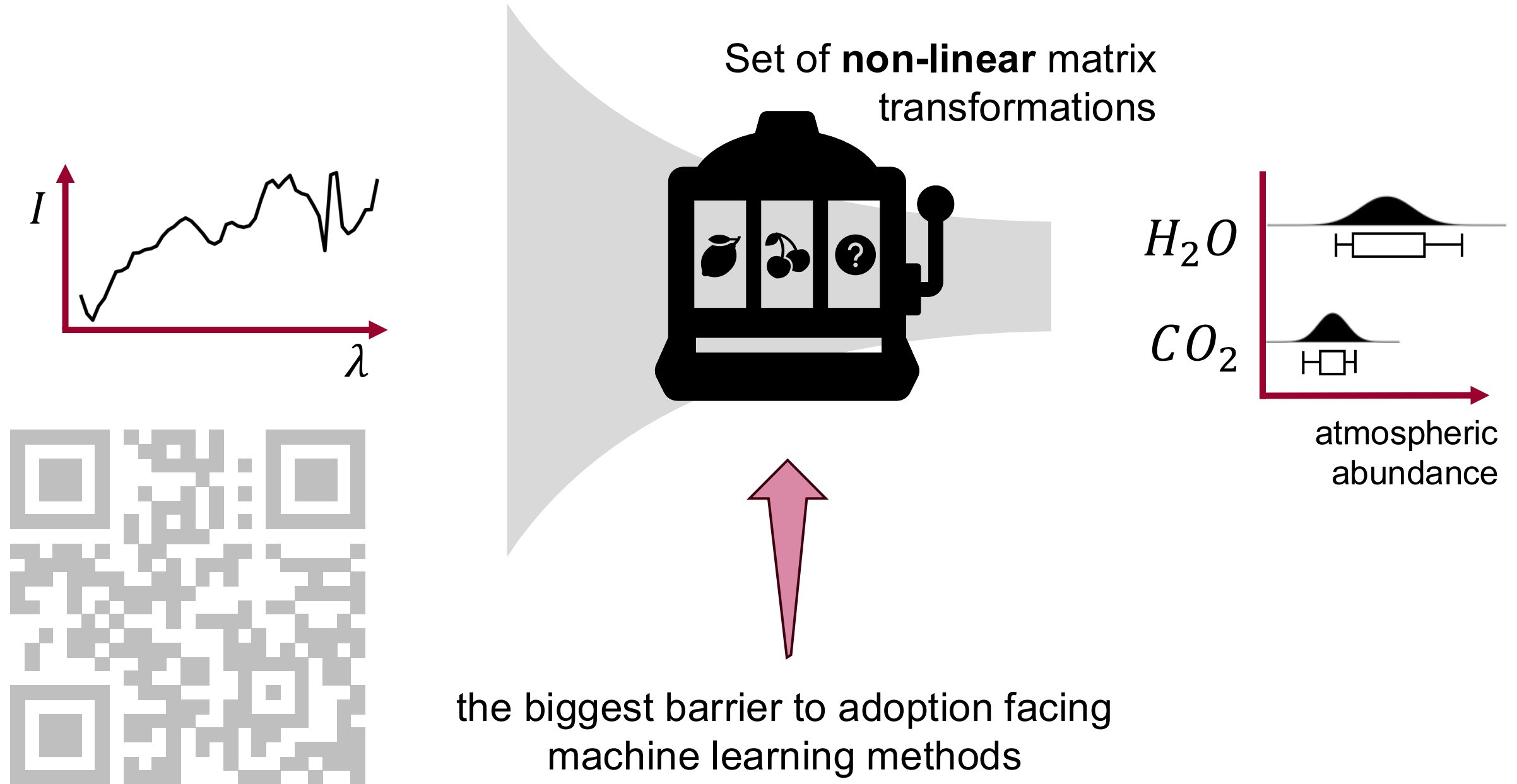
Interpreting and Validating Interpretability Metrics

Jools Clarke

Contact me: joolsclarke.co.uk / j.d.clarke@ucl.ac.uk

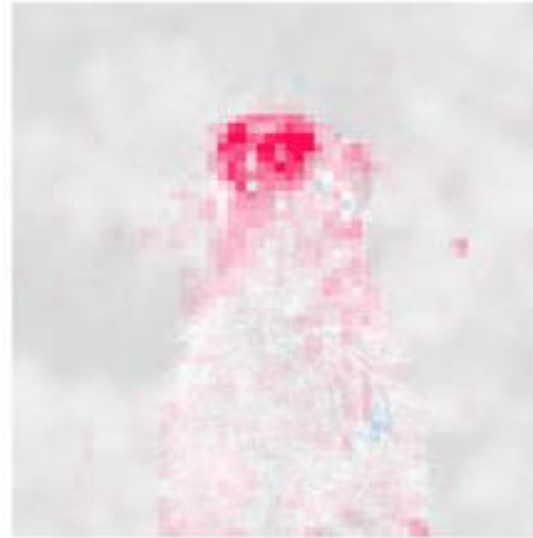


Download these
slides

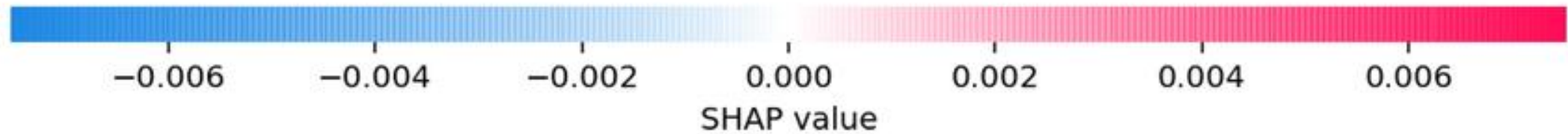
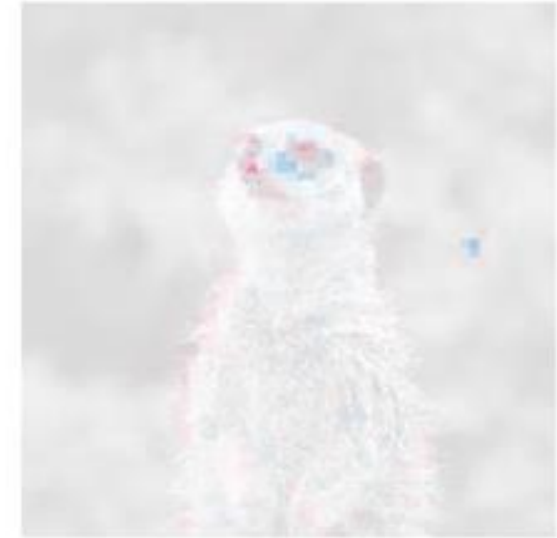




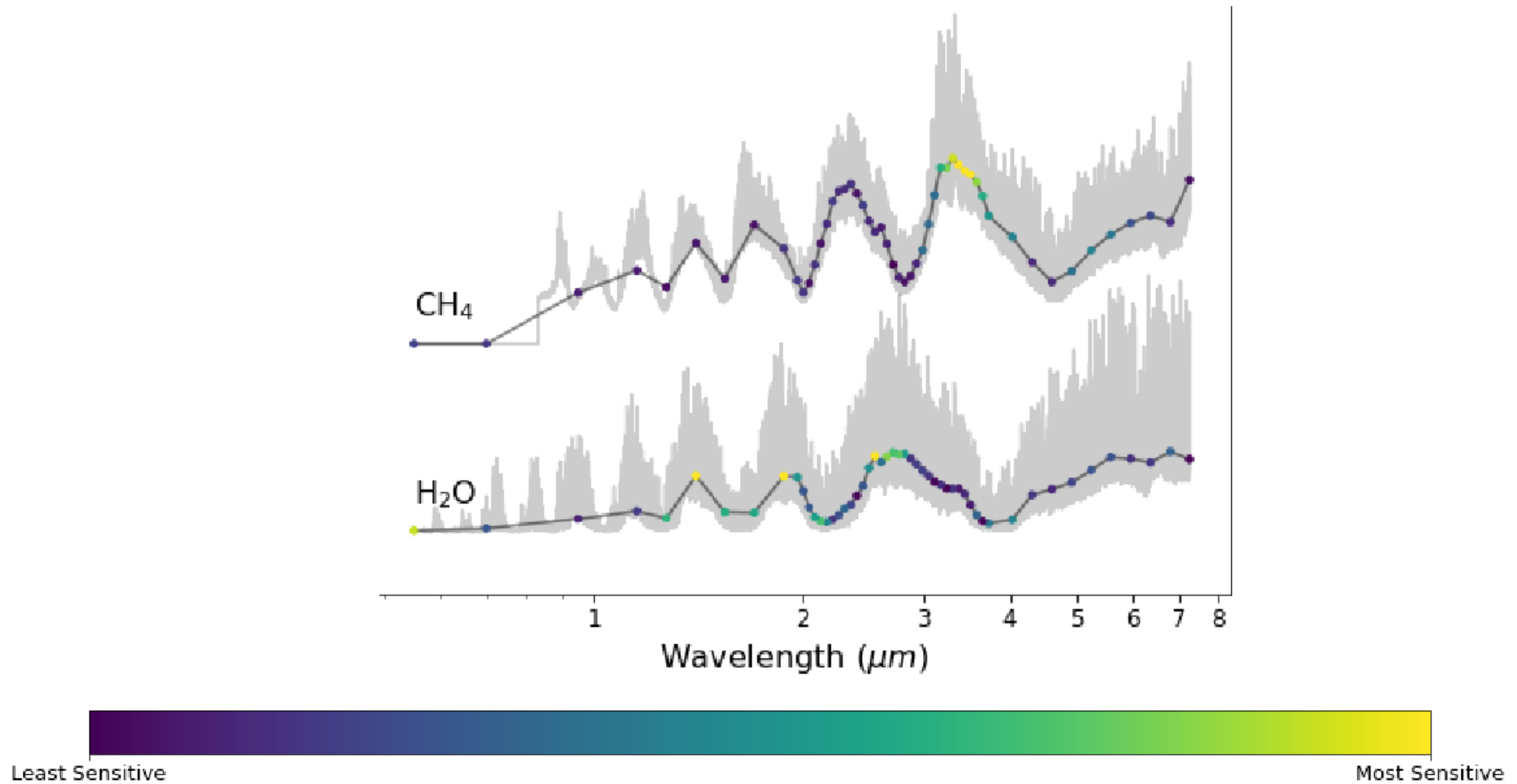
meerkat



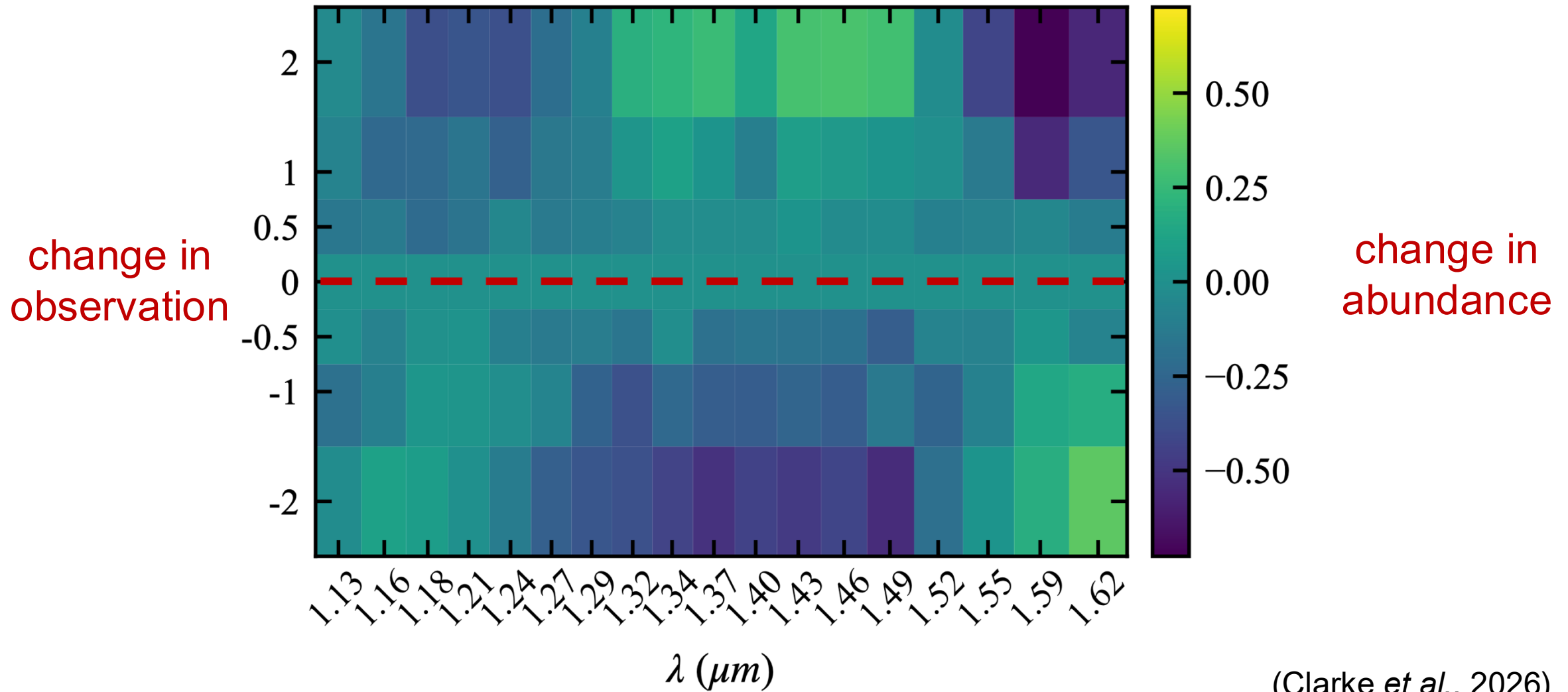
mongoose

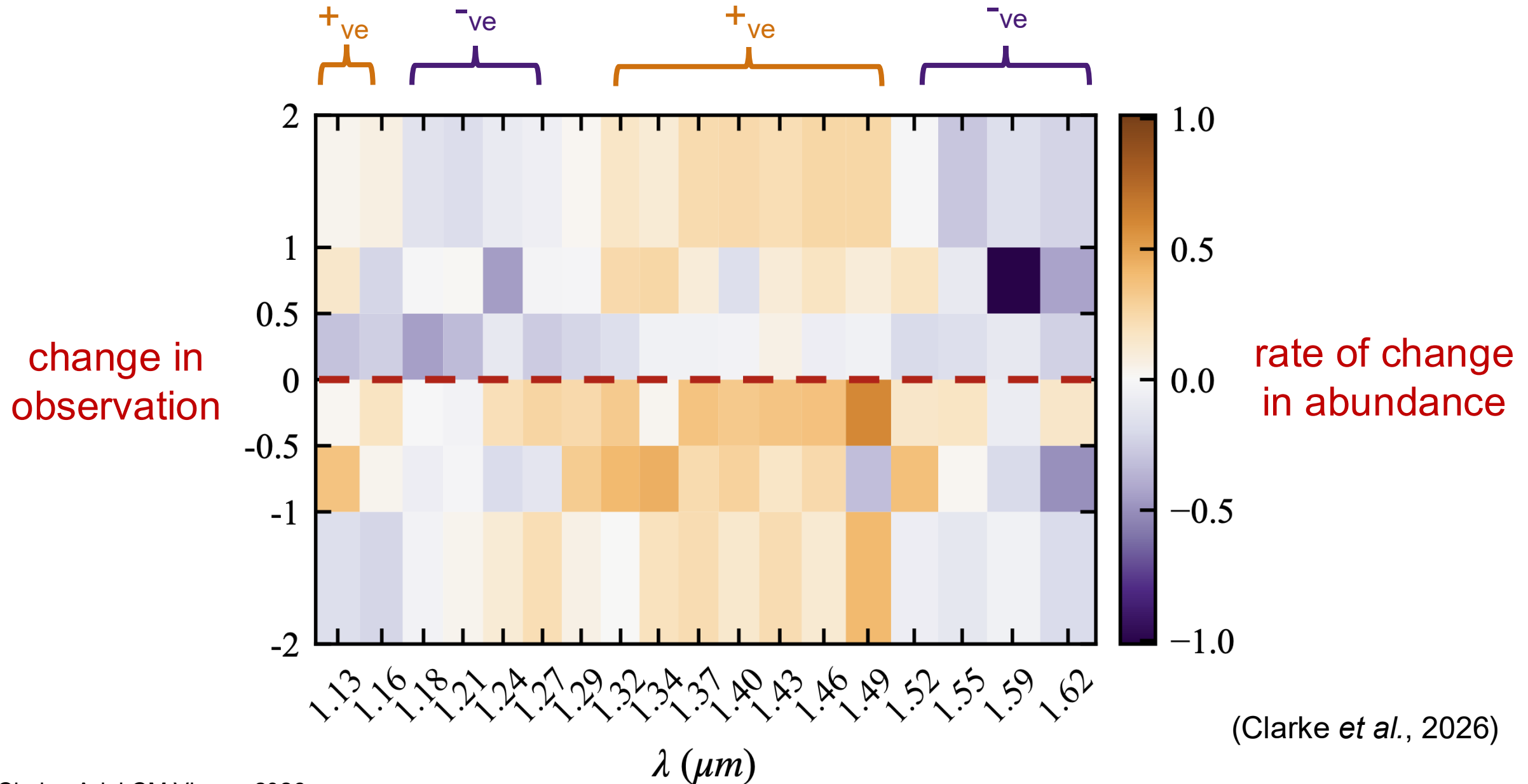


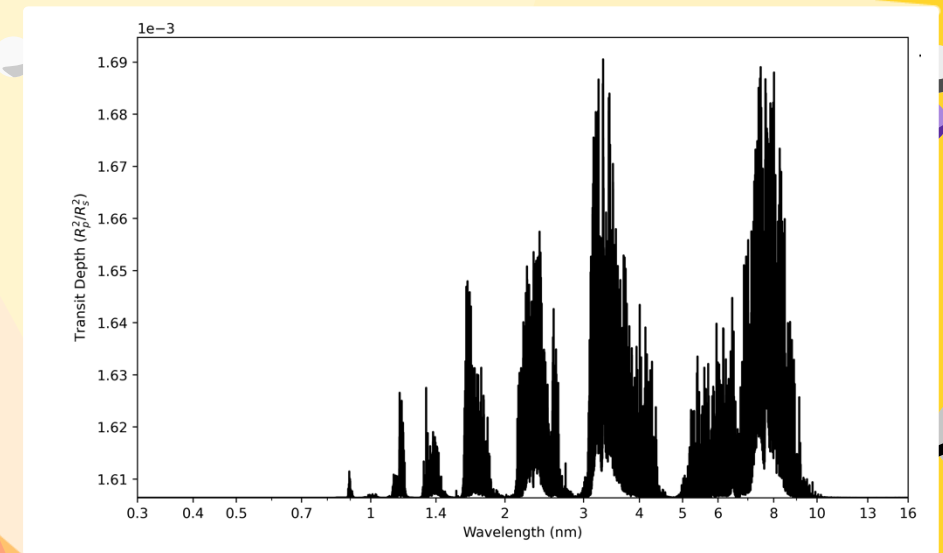
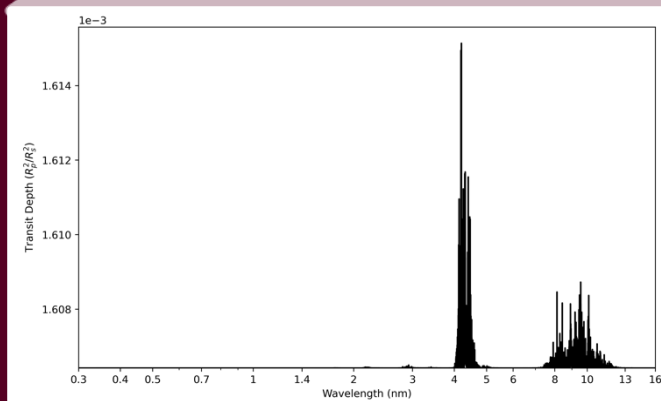
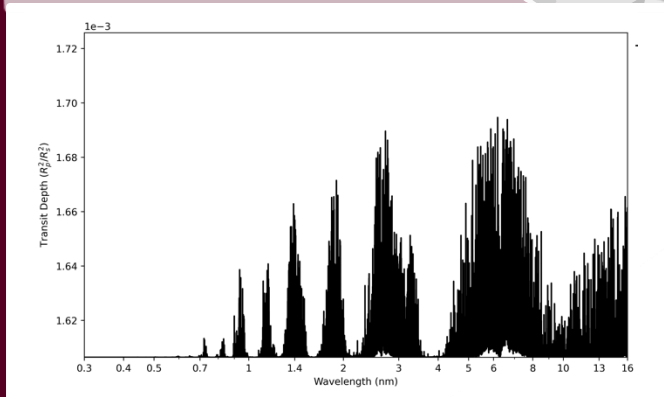
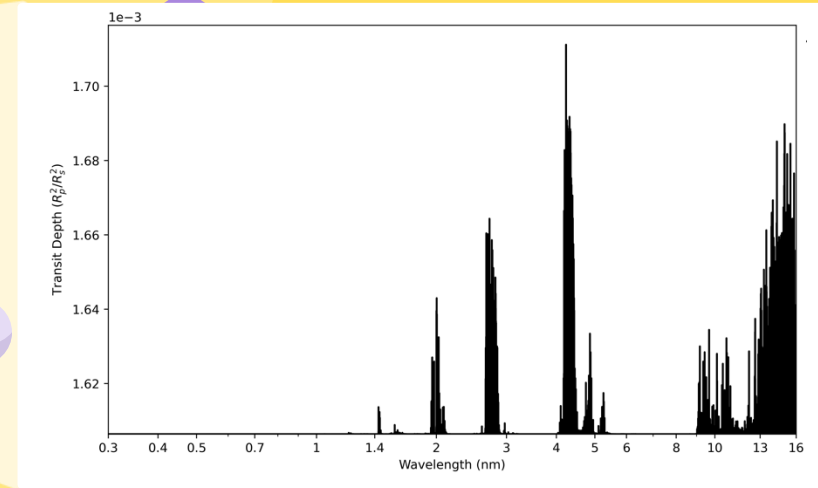
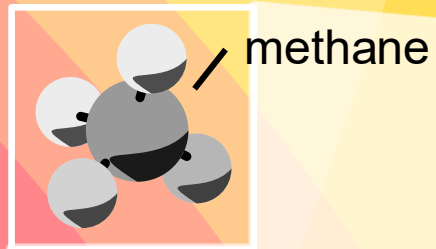
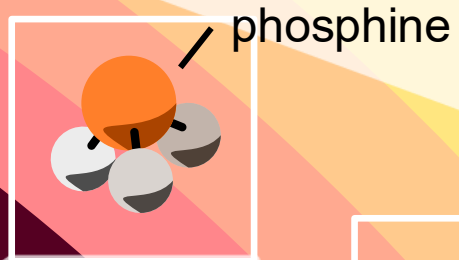
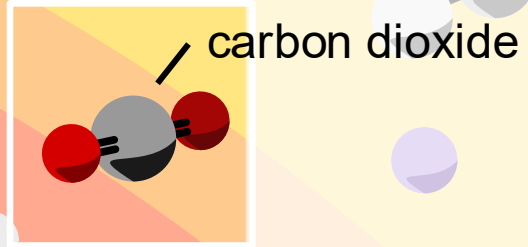
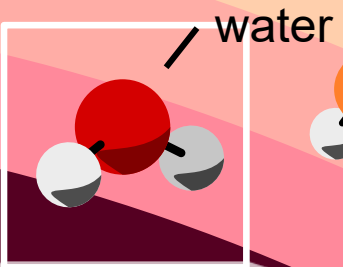
SHAP paper: (Lundberg and Lee, 2017), image: (Shojaeinasab *et al.*, 2024)

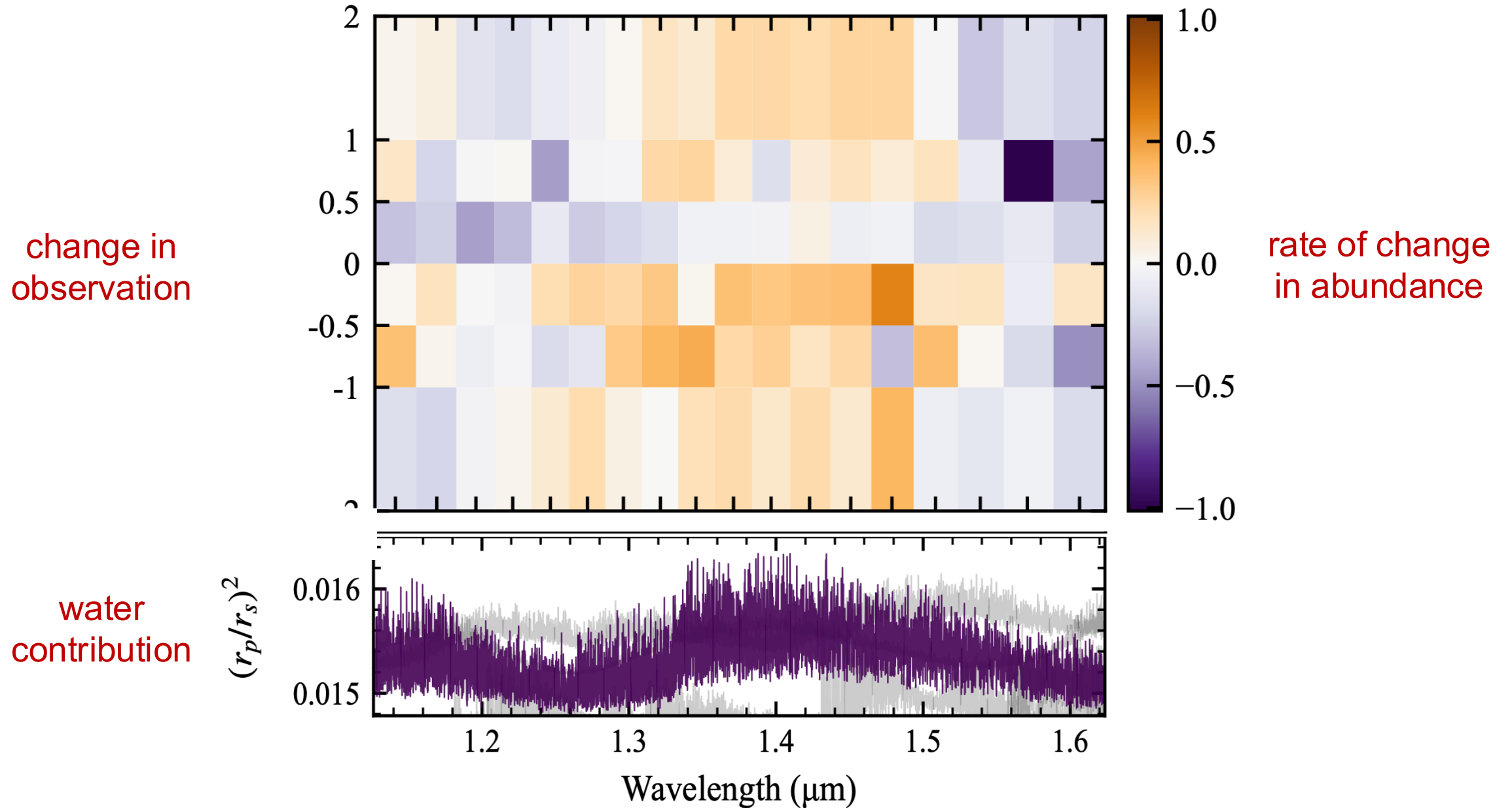


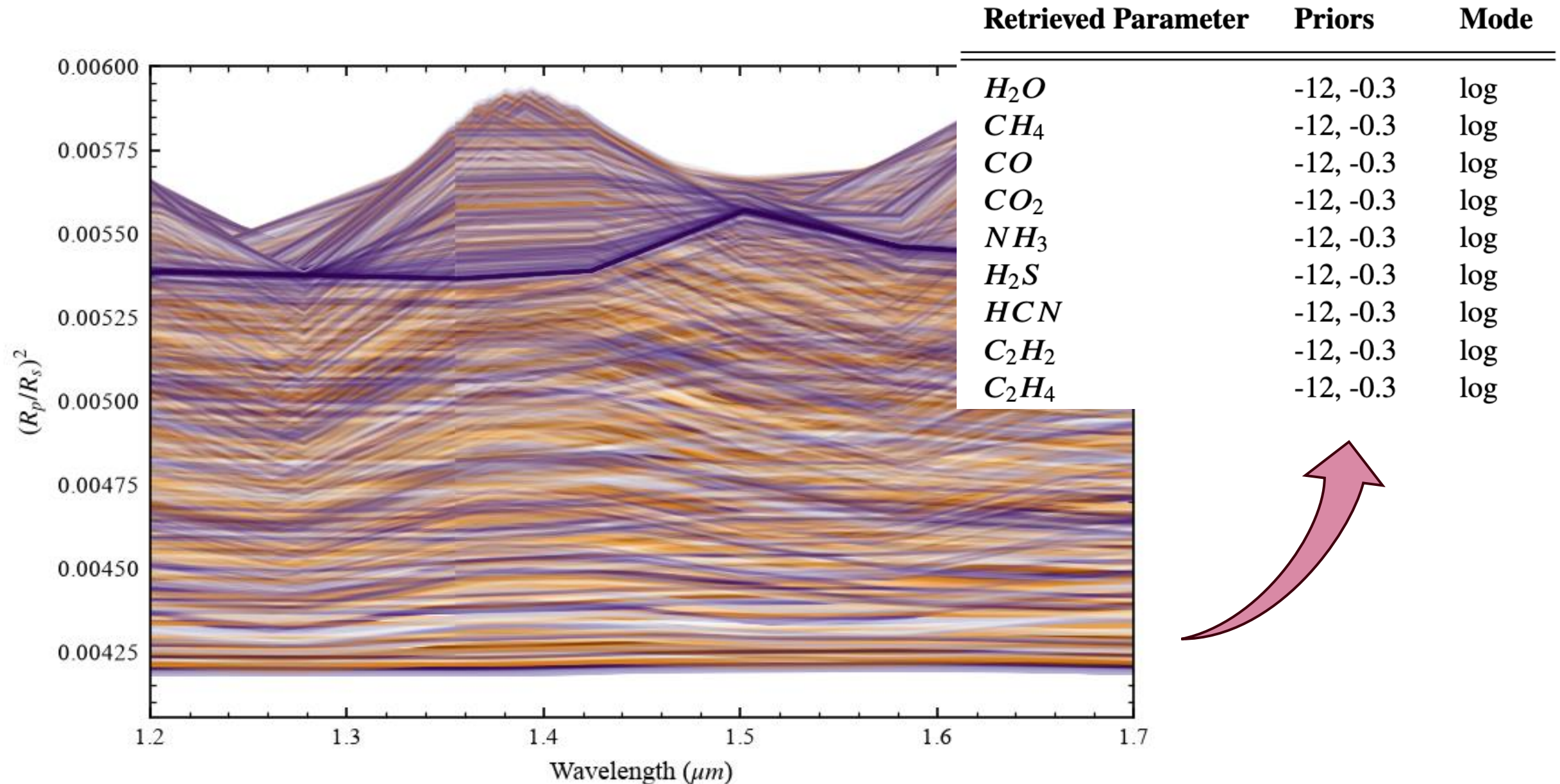
(Yip *et al.*, 2021)

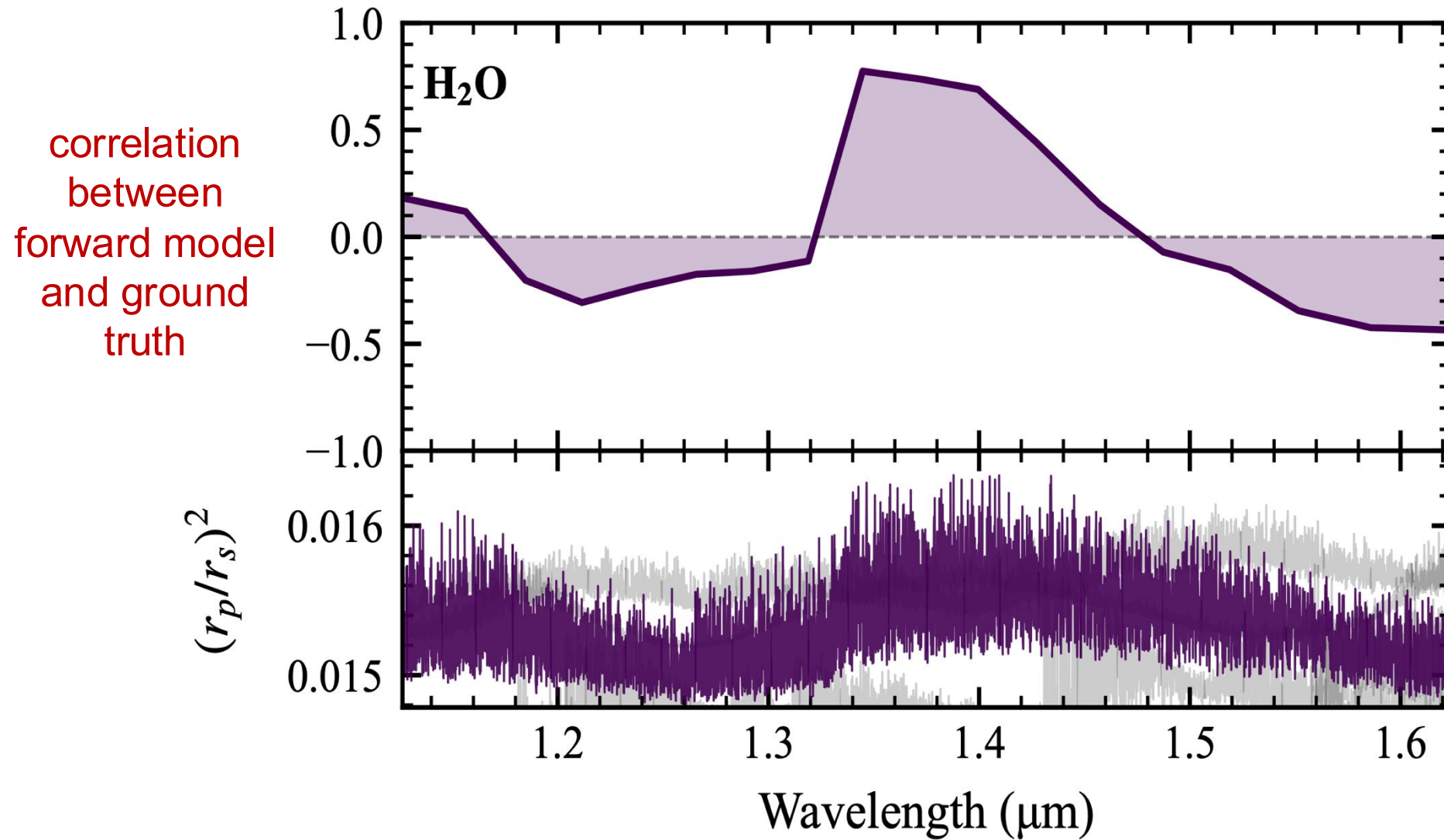
(Clarke *et al.*, 2026)

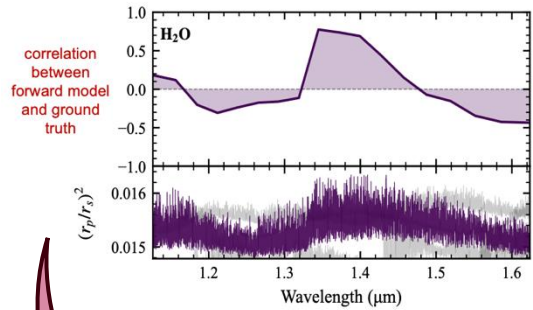




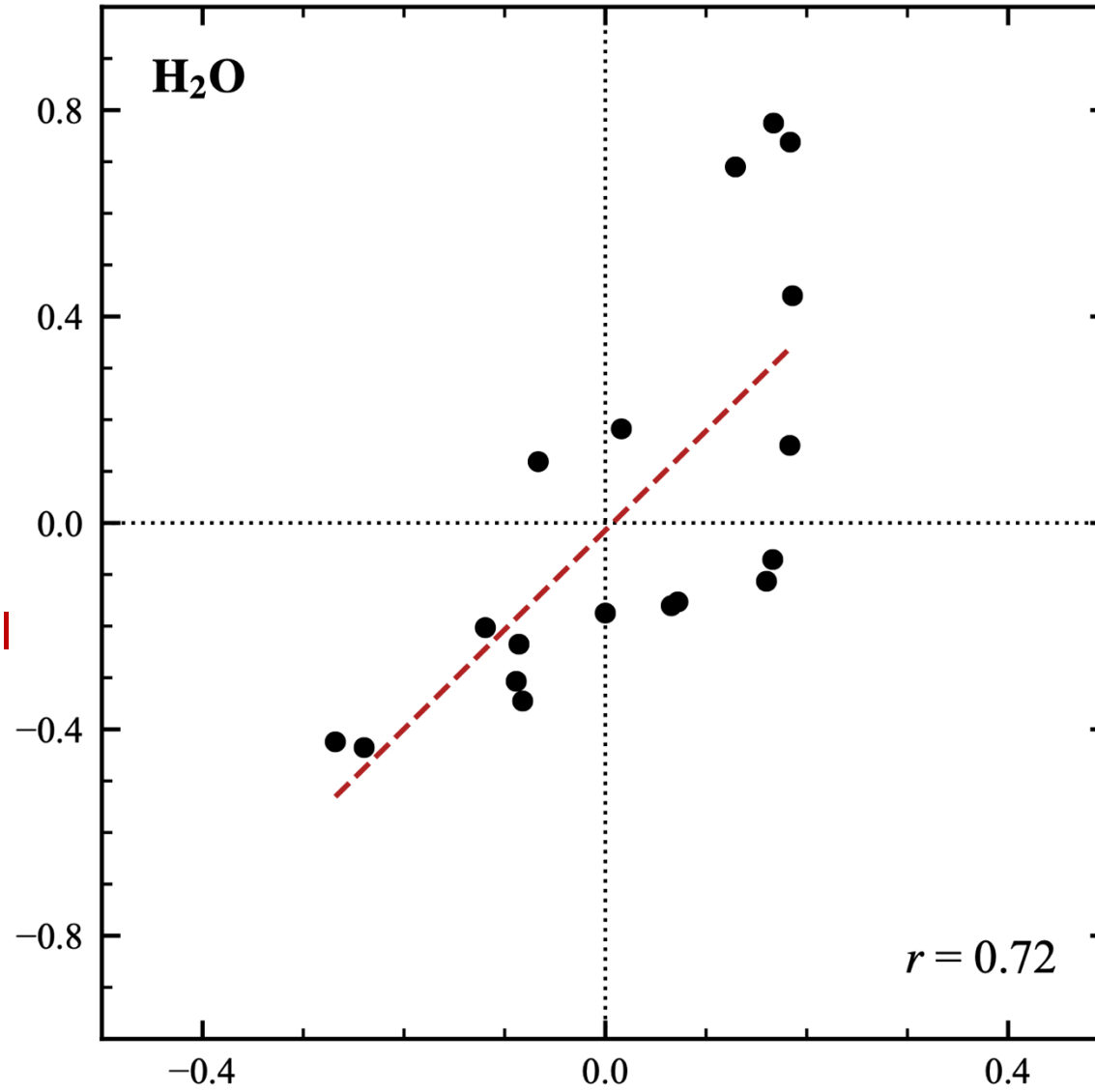




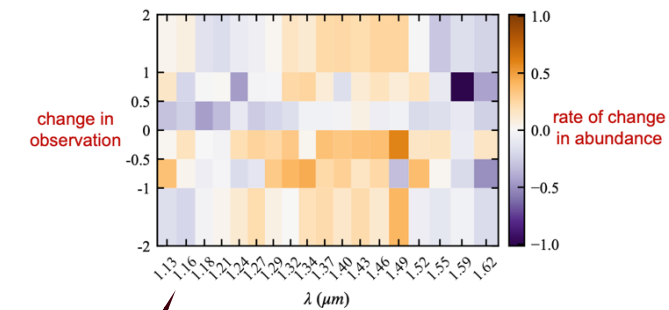




correlation between forward model and ground truth

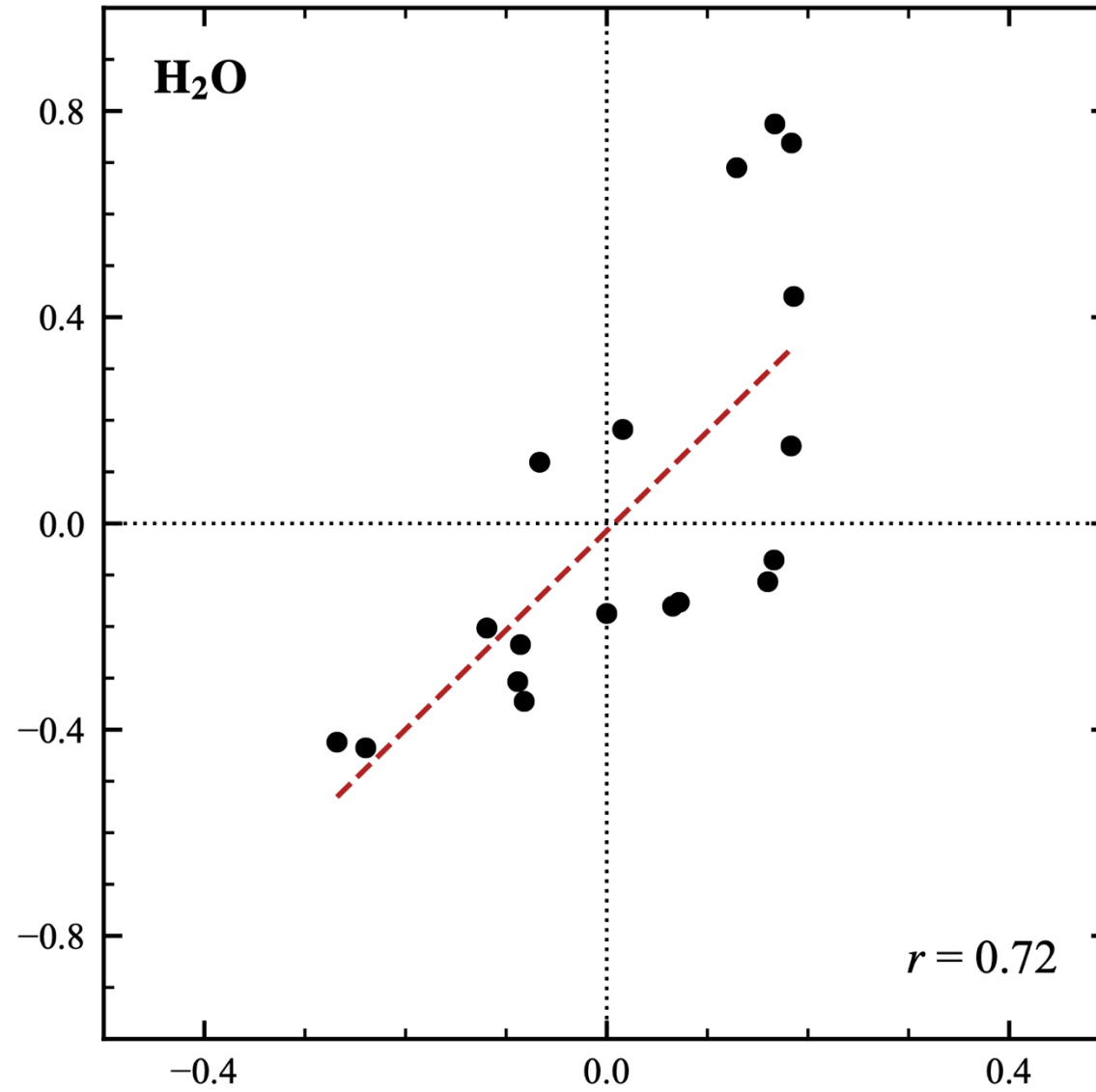


1 point
=
1 wavelength



average rate of change in abundance

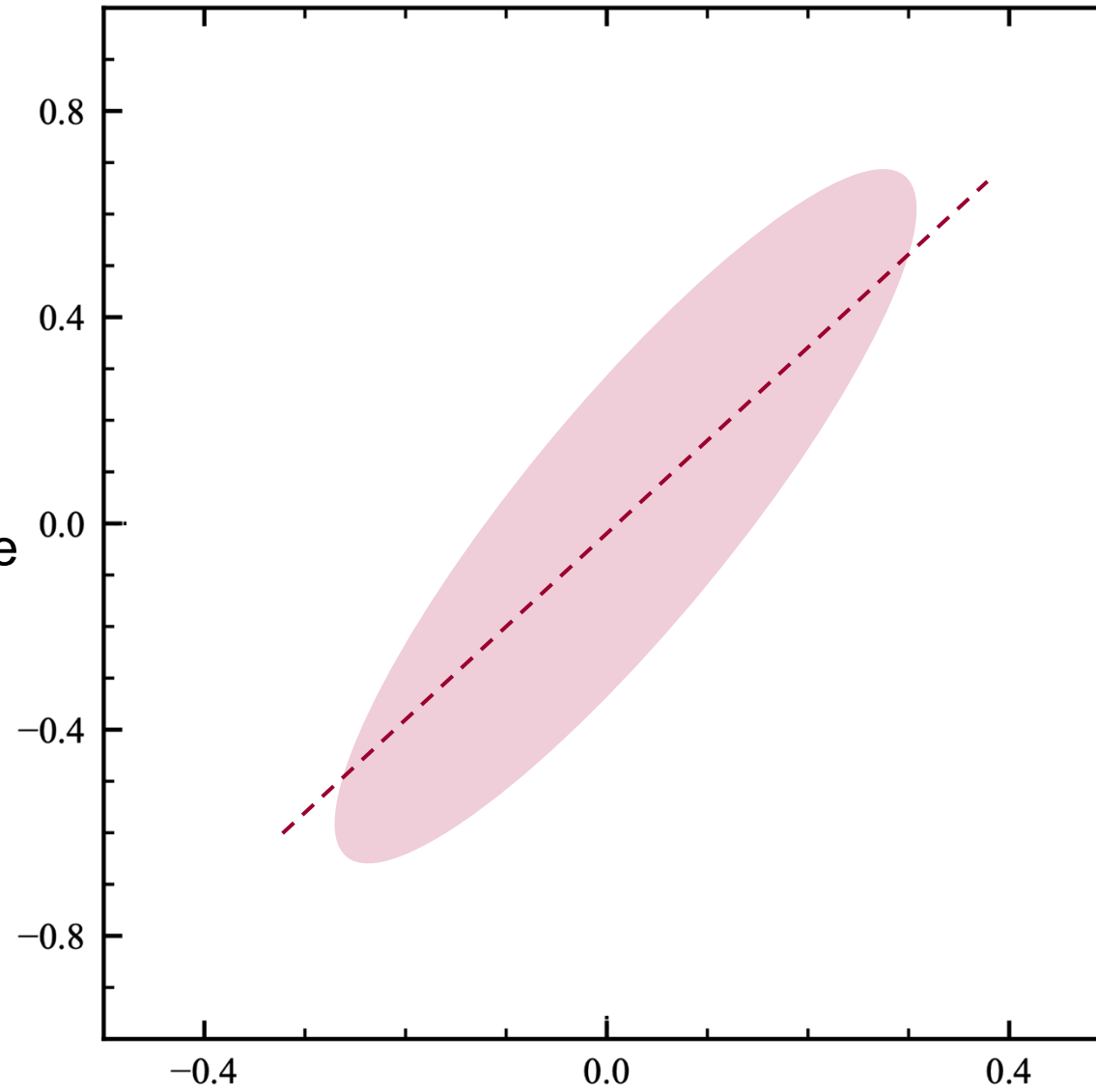
correlation between
forward model
produced by **TauREx-III**
and ground truth



What is a good metric?

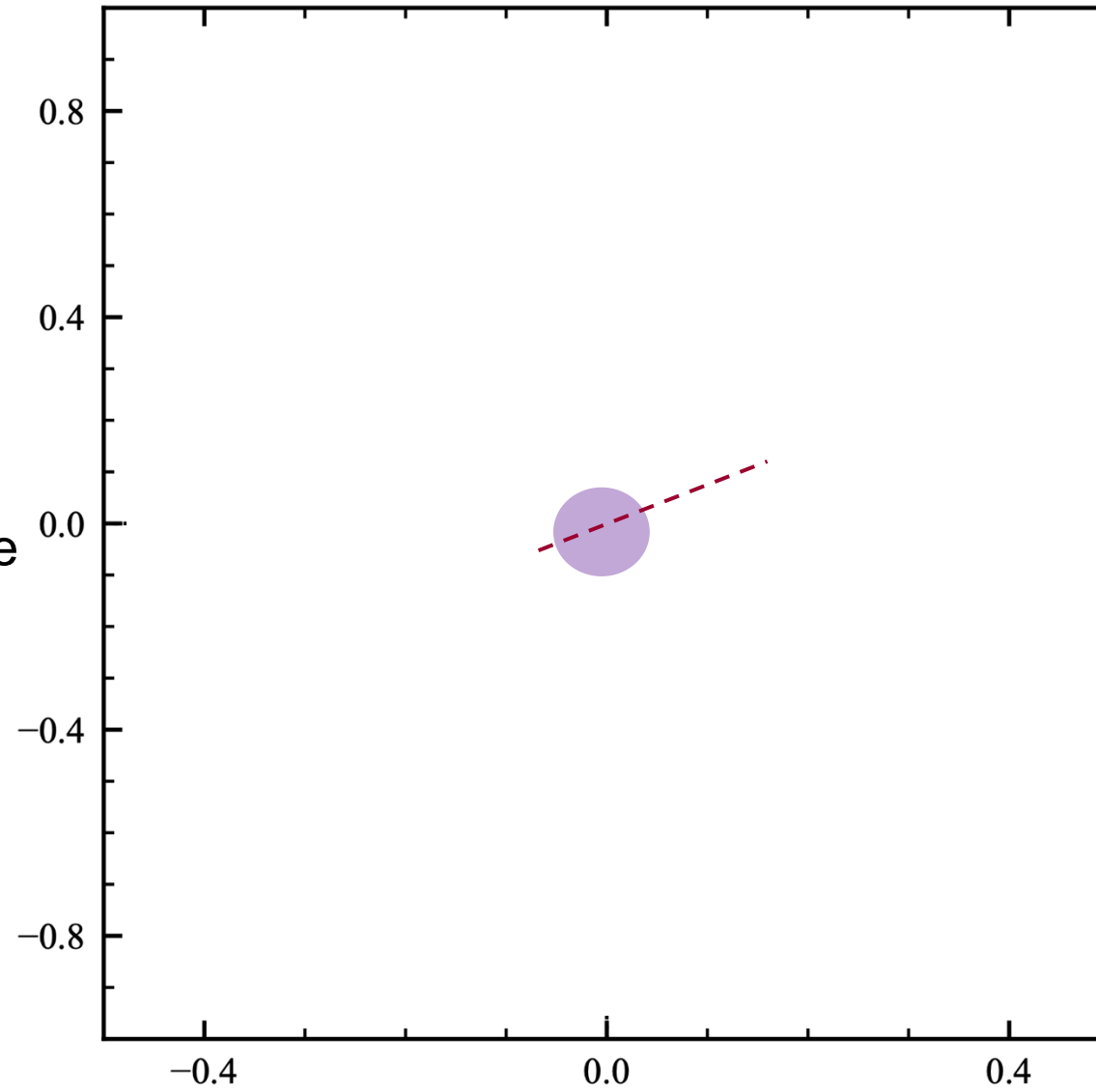
average rate of change in mean posterior
abundance **retrieved by TauREx-III**

average rate of change in
mean posterior abundance
retrieved by ML surrogate

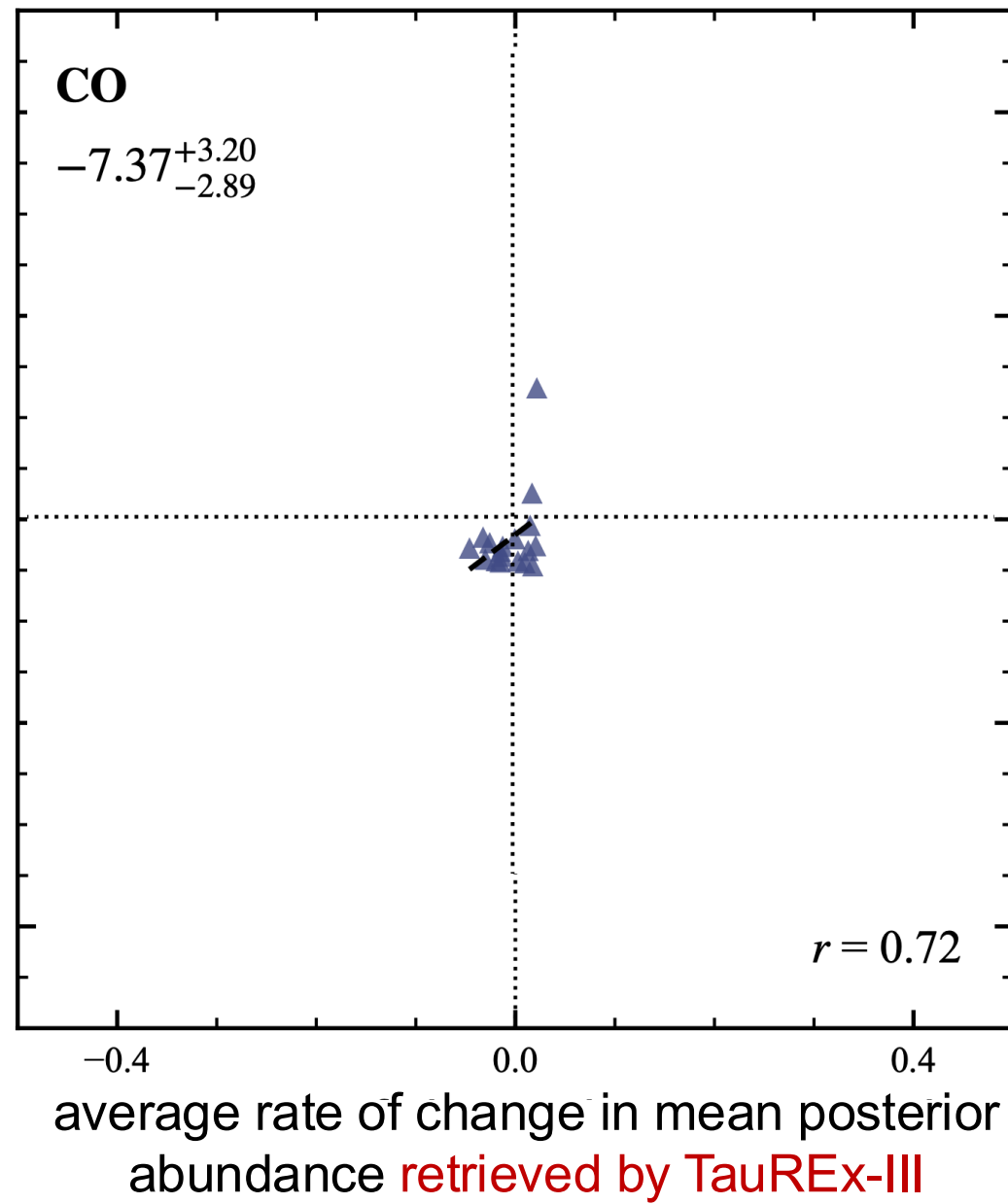


The ML surrogate
interprets the data
in *the same way* as
TauREx.

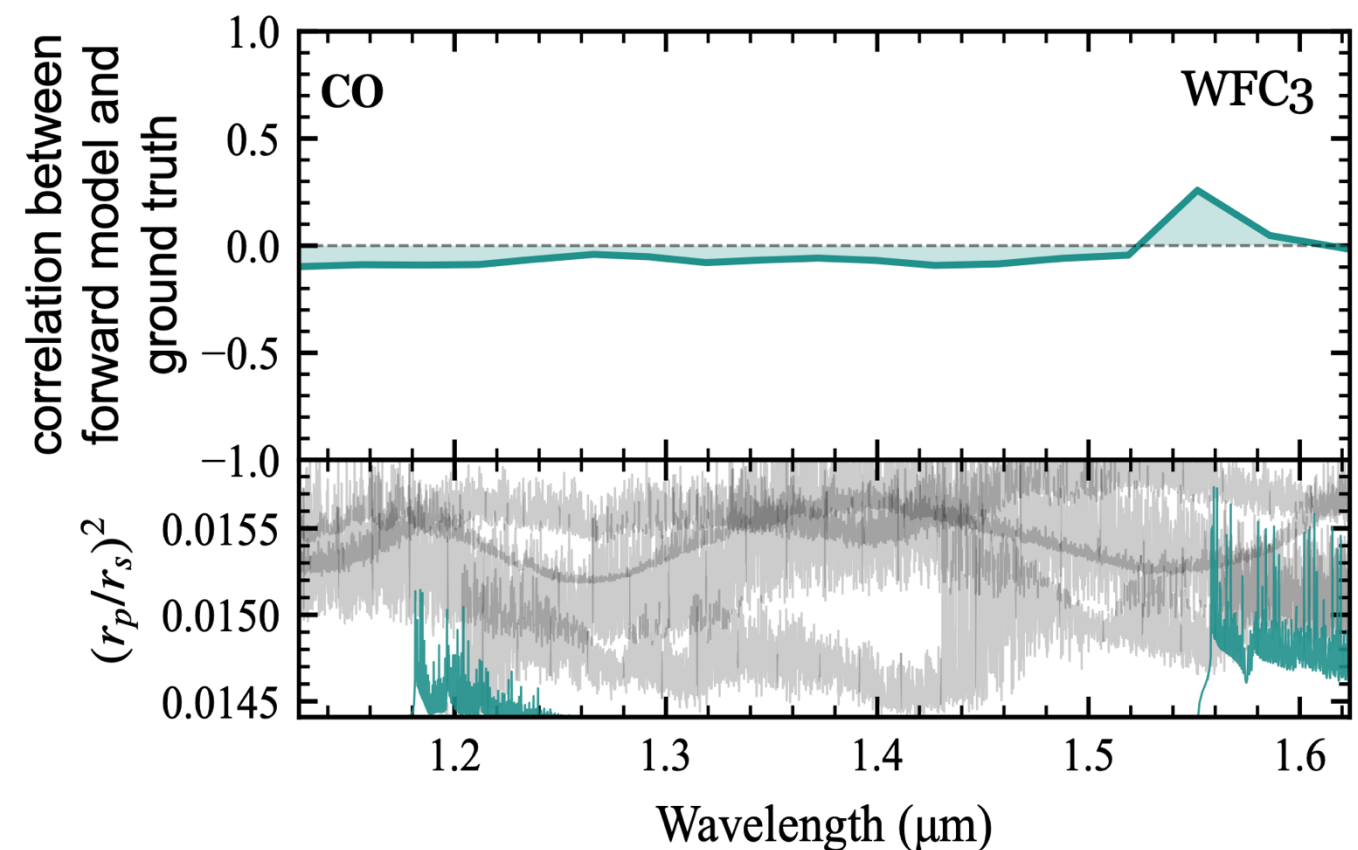
average rate of change in
mean posterior abundance
retrieved by ML surrogate



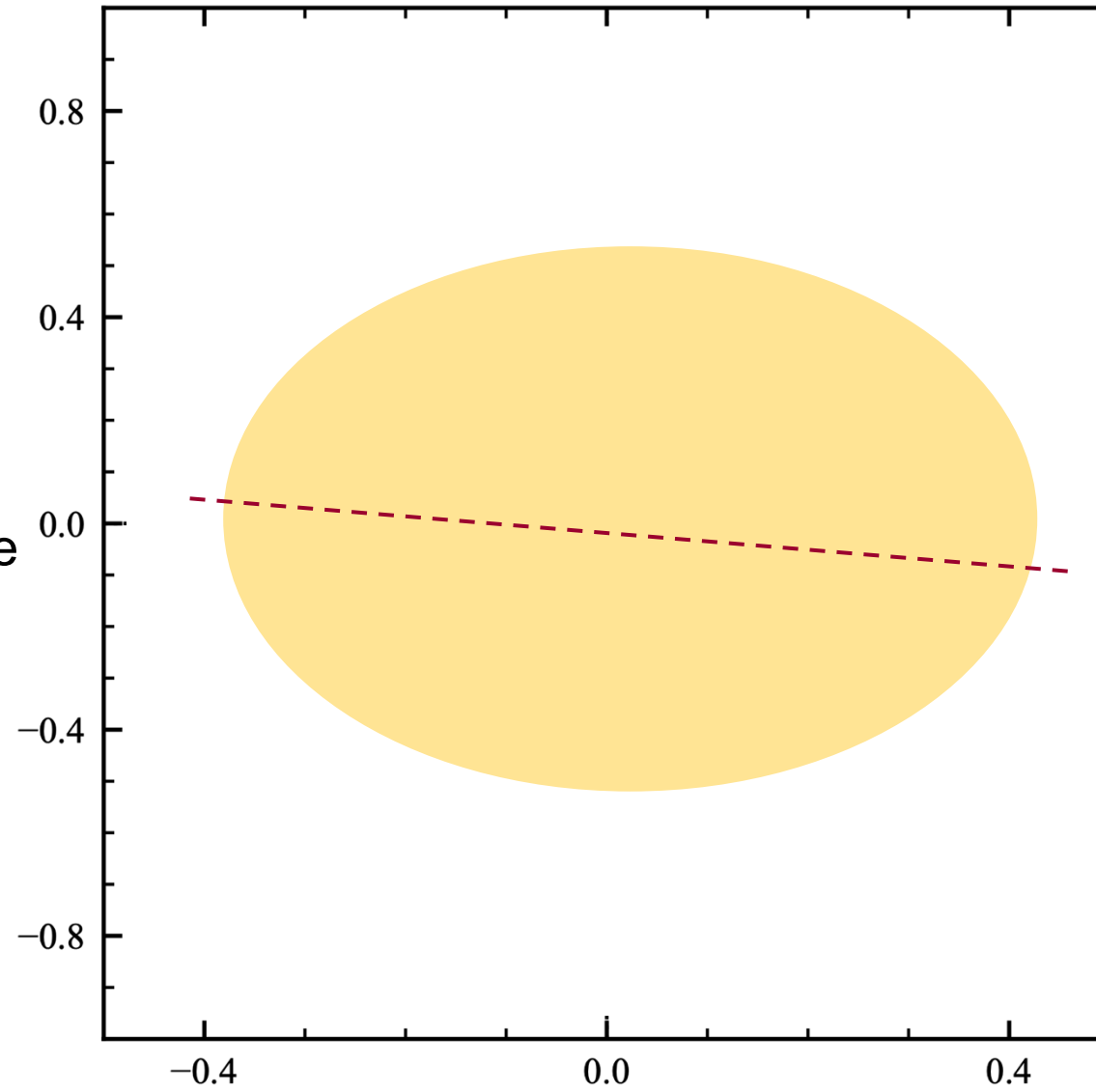
**The data does not
constrain the
solution well.**



The data does not constrain the abundance well.



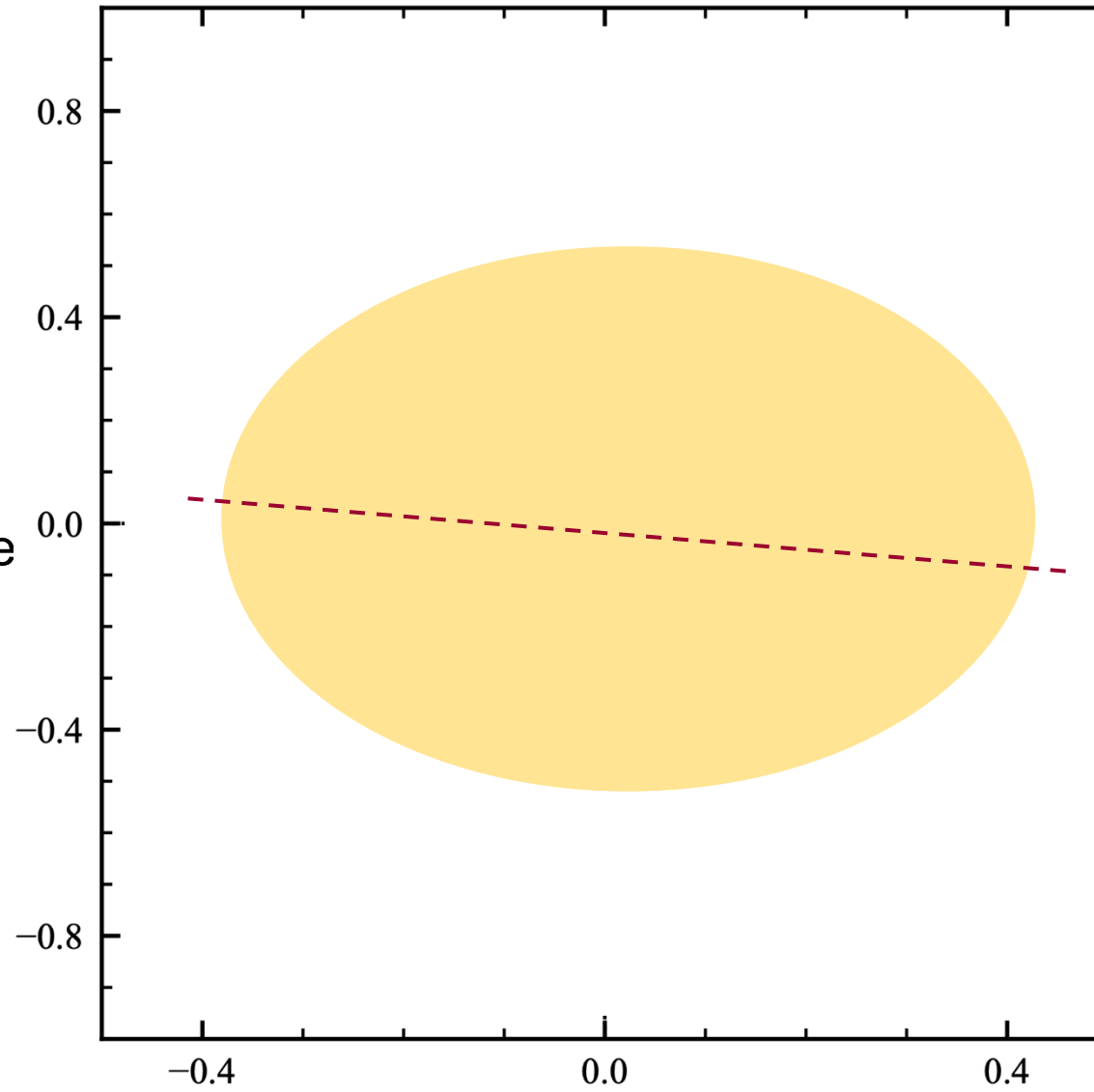
average rate of change in
mean posterior abundance
retrieved by ML surrogate



The ML surrogate
interprets the data
differently to
TauREx.

This is *not* the same
as low model
accuracy.

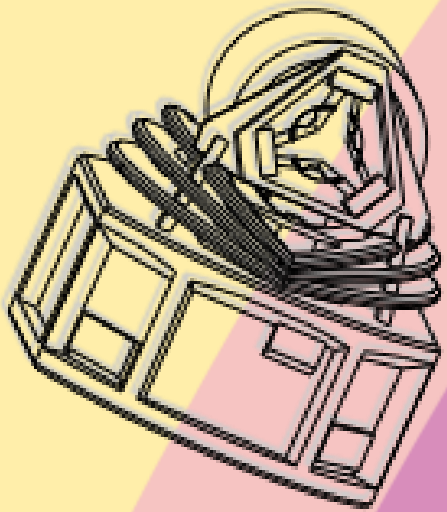
average rate of change in
mean posterior abundance
retrieved by



**Poor model
optimisation?**

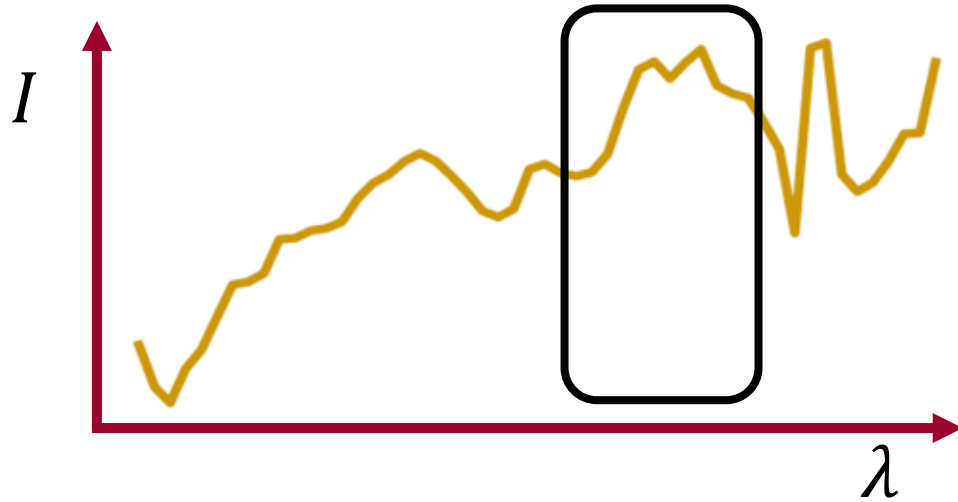
**New ways to break
degeneracy?**

Machine Learning Retrievals for Practical Application



Wish List

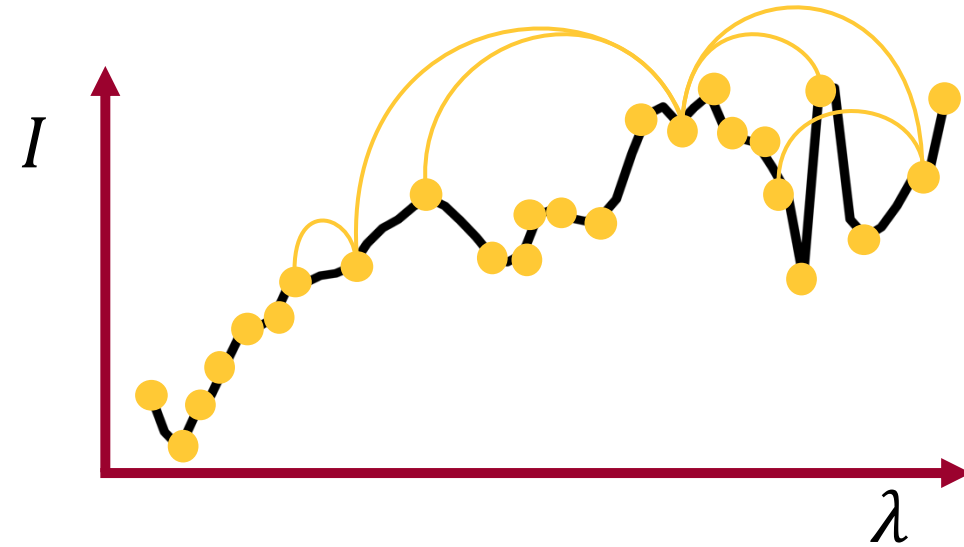
- ✓ *uncertainty quantification*
- ✓ *feature importance*
- ? *wavelength grid / instrument agnostic*
- ✗ *joint posterior*

CNN

$$\begin{bmatrix} I_1 \\ I_2 \\ \vdots \\ I_N \end{bmatrix}$$

One large
N-dimensional
datapoint

**sampling grid is baked into the
model as the true function**

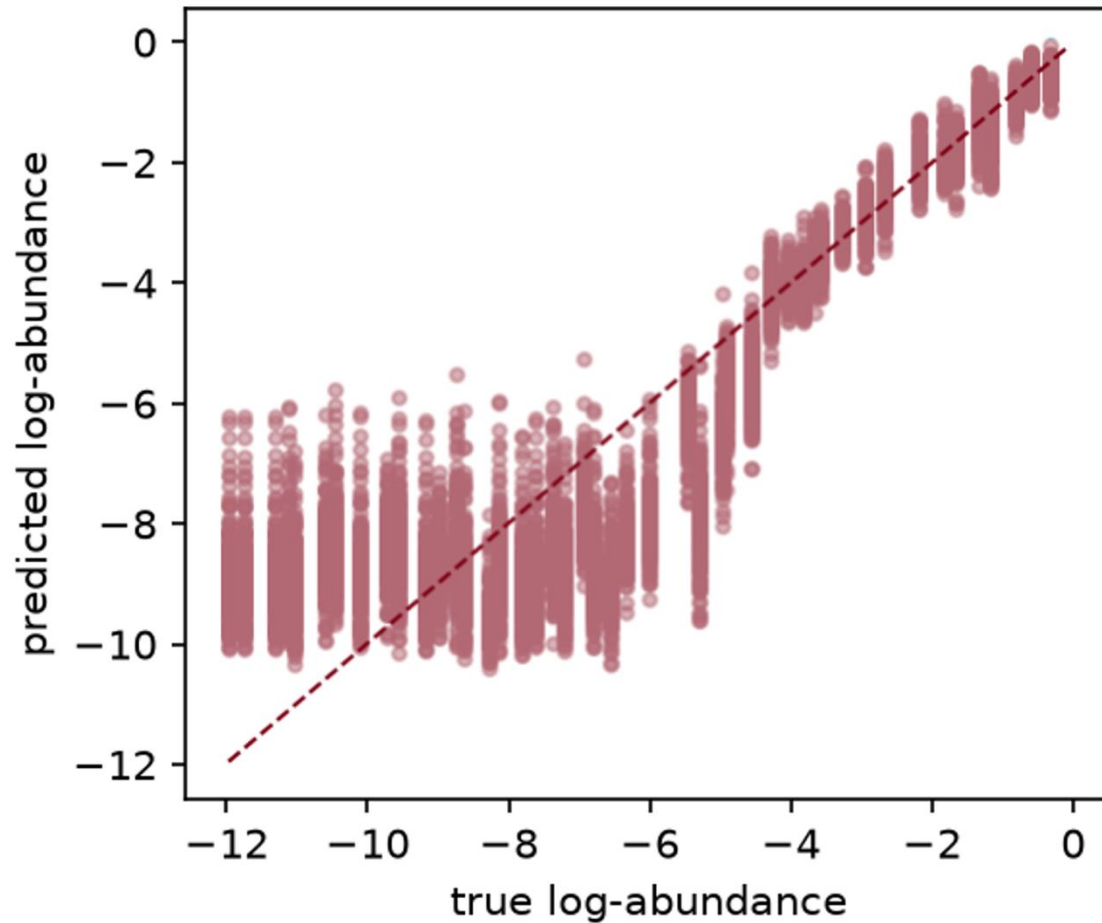
Set Transformer

$$\begin{bmatrix} I \\ \lambda \\ \text{width}_\lambda \end{bmatrix}$$

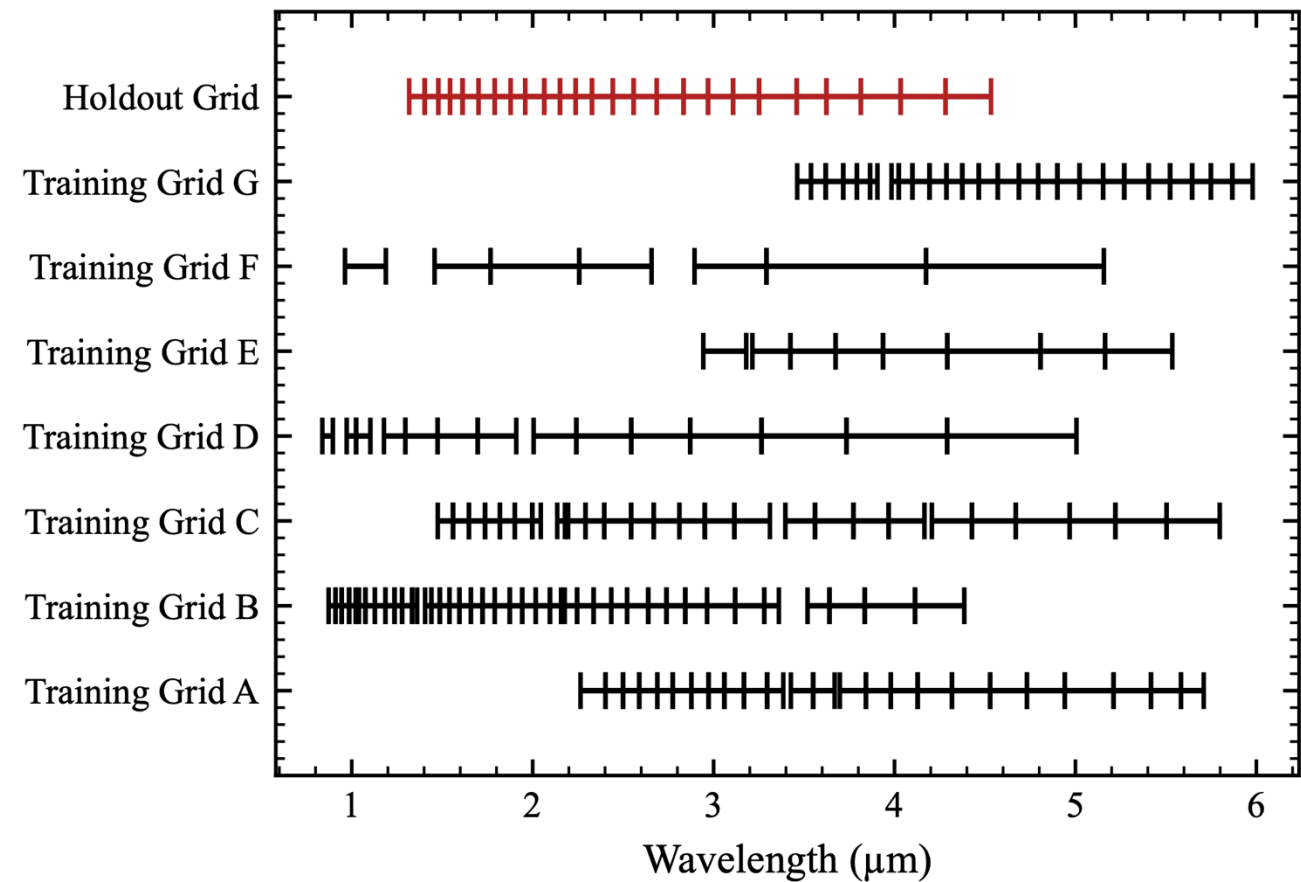
A set of N
3-dimensional
datapoints

**datapoints provide information on
an unknown true function**

Preliminary methane abundance retrievals for unseen wavelength grid.



Training across different data reductions and different instruments.



**Get in touch, get updates for this work,
and for any of my other research here!**



**I have less than a year left
on my PhD...**

